

Continuous vs. Discrete

TEACHER KEY | Algebra II Honors | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can classify a domain as continuous or discrete, write each with the notation that fits, recognize step behavior, and judge when discreteness is a property of the quantity versus a modeling choice.

Vocabulary

Continuous — every value between any two points belongs to the domain. **Discrete** — the domain is **separated points**.

Step function — a **continuous** domain paired with a **discrete** range. The output jumps.

Sequence — a function whose domain is the **counting numbers**. Always discrete.

Part 1 — Classify and Justify

Write C or D, then give the domain in correct notation and one sentence of justification.

Situation	C/D	Domain	Why
Distance a car travels in 3 hours	C	$[0, 3]$	time flows
Number of students on 6 buses	D	$\{0, 1, \dots\}$	whole people
Mass of water in a beaker	C	$[0, \text{max}]$	any amount
Term number in a sequence	D	$\{1, 2, 3, \dots\}$	counting only

Part 2 — One Equation, Two Domains

A cube-shaped box of side x has volume $V(x) = x^3$.

x	1	2	3	4
V(x)	1	8	27	64

Any size box: domain $(0, \infty)$ range $(0, \infty)$ **continuous**

Built from 1-inch blocks: domain $\{1, 2, 3, \dots\}$ range $\{1, 8, 27, \dots\}$ **discrete**

The algebra did not change. What did? **the situation being modeled**

Part 3 — Step Behavior

A parking garage charges \$3 for each hour or any part of an hour, up to 4 hours. Park for 1.2 hours and you pay for 2.

t (hours)	0.5	1.0	1.2	2.0	2.7	3.5
Cost	\$3	\$3	\$6	\$6	\$9	\$12

Domain (time parked): $(0, 4]$ **continuous** Range (cost): $\{3, 6, 9, 12\}$ **discrete**

This is the case the on-level notes do not cover: a domain and range that disagree. Explain how that happens.

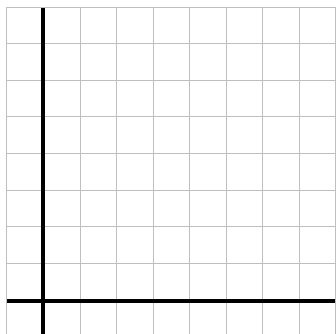
Time varies smoothly, but the **price jumps** at each whole hour instead of rising gradually.

Why must $t = 1.0$ cost \$3 and not \$6? **exactly one hour is not "part of a second hour"**

Part 4 — Graph Both Kinds

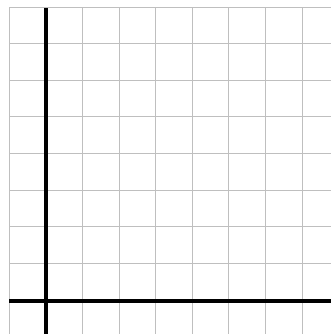
Each square is 1 unit. Use open and closed dots carefully.

- A. Discrete.** A tutor charges \$2 per session, up to 4 sessions. **B. Continuous.** A candle burns 2 cm per hour from 8 cm.



Domain **$\{1, 2, 3, 4\}$**

Range **$\{2, 4, 6, 8\}$**



Rule $h = 8 - 2t$

Domain **$[0, 4]$** Range **$[0, 8]$**

Part 5 — Going Further

1. Is money continuous or discrete? Argue both sides, then commit to one.

Discrete in practice (**smallest unit is a cent**), often modeled as continuous because the steps are tiny.

2. Give a situation with a discrete domain and a continuous range, or argue it is impossible.

Impossible — **finitely many inputs can only produce that many outputs**

Exit Ticket

Postage costs \$0.60 for the first ounce and \$0.24 for each additional ounce or part of an ounce, up to 5 ounces.

Domain (weight) **$(0, 5]$** Range (cost) **$\{0.60, 0.84, 1.08, 1.32, 1.56\}$**

Classify the domain and the range, and explain why they differ: **continuous domain, discrete range - a step function**